CIRCL
Computer Incident
Response Center
Luxembourg

Drone Threat Intelligence Workshop - CTI

FIRST 2026 - Cyber Threat Intelligence Conference

Paul Jung - paul.jung@circl.lu

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CIRCL https://www.circl.lu

TLP:CLEAR

CIRCL as a NIS2 CSIRT and Open Source Security Leader

- **CIRCL is Luxembourg's national CSIRT for the economy**, designated under the NIS2 Directive.
- Core missions: incident handling, early warning, threat analysis, coordinated vulnerability disclosure, coordinated response, and support to essential/important entities.
- **Open source is central to our mandate:** CIRCL develops and maintains **17+ security projects** used worldwide by CSIRTs, ISACs, industries, intelligence community and defenders.
- Key platforms: **MISP**, **AIL**, **Vulnerability-Lookup**, **FlowIntel**, **LookyLoo**, **Cerebrate**, **GCVE.eu**.
- Open source ensures **transparency, interoperability, sovereignty** and long-term sustainability for operational security.

How to get UAV OSINT Related Information

- **Vendor and manufacturer websites**
  - Commercial and military Website such as Aerosentinel, Thales, etc...
- **Drone and defense exhibitions**
  - Sources such as Dronexpo, Eurosatory, and other defense trade fairs
  - `https://www.eventseye.com/fairs/st0_salons_defense-armement.html`
  - X is also a good source. (`@TKouilou`, `@OpexNews`, `@AlexXplore`)
- **Incident listing Websites**
  - `https://dronedj.com/guides/drone-incident/`
  - `https://d-fendsolutions.com/drone-incident-tracker/`
  - `https://map.faineg.com/public/index.html` (Attacks on Civilans)
  - `https://en.wikipedia.org/wiki/List_of_unmanned_aerial_vehicle-related_incidents` (less regularly updated)

How to get UAV-Related Information

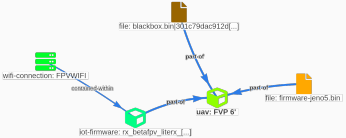
- **Telegram Channels**
  - Many drones-related channels
  - Many material originating from UA / Russia War.
  - Access to resources such as hacks, catalogs, technical information, and tactical guides
- **In case you need a tool**
  - `https://www.ail-project.org/`
  - Archiving, pivoting, translation, and PDF translation

How Can a Drone Incident Be Addressed?

- How to share information efficiently ?
- How to keep traceability of key elements ?
- How to correlate and cross-check findings ?

Drone Incident – Data Modeling and Information Exchange

- **Group / Coordination Information:**
  - Agencies involved (law enforcement, aviation authority, military)
  - Related incidents or known threats
- **Incident Information:**
  - Date, time, and geographic location
  - Type of drone (commercial, FPV, military)
  - Observed behaviour (reconnaissance, loitering, targeting)
  - Impact on infrastructure or airspace
- **Forensic Information:**
  - Serial numbers and hardware identifiers
  - Flight controller (FC) configuration
  - Flight logs / Blackbox data
  - Observed radio frequencies
  - Identifiers (UID, Remote ID, pilot/aircraft names if present)



Drone modeling

MISP

What is MISP ??

MISP Objects

Structured drone artefacts for sharing, enrichment and correlation

MISP Drone Object - UAV

Object name: `uav`

Link: `https://github.com/MISP/misp-objects/blob/main/objects/uav/definition.json`

- Identifies the drone with manufacturer, model, variant and serial number.
- Captures the mission context: operator, mission type/intent, payload and country of operation.
- Stores technical traces such as firmware version/hash, controller ID, MAC address and flight path.

MISP Drone Object - Controller

Object name: `remote-controller`

Link: `https://github.com/MISP/misp-objects/blob/main/objects/remote-controller/definition.json`

- Describes the controller identity with manufacturer, model and serial number.
- Tracks technical identifiers such as controller ID, MAC address, IMEI and FCC ID.
- Records the controller firmware version/hash and optional picture or manufacturing year.

MISP Drone Object - Firmware

Object name: `iot-firmware`

Link:
`https://github.com/MISP/misp-objects/blob/main/objects/iot-firmware/definition.json`

- Stores the firmware sample itself with filename, version and file format.
- Supports integrity and correlation with hashes such as SHA256, SHA512, SHA1 and MD5.
- Preserves analysis artefacts including binwalk output, entropy graph and boot logs.

MISP Drone Object - Drone RID Observation

Object name: `drone-rid-observation`

Link: `https://github.com/MISP/misp-objects/blob/main/objects/drone-rid-observation/definition.json`

- Captures the observation timestamp together with UAS ID, operator ID or session ID.
- Records flight and control-station telemetry: latitude, longitude, altitude, speed, heading and operator position.
- Keeps broadcast context such as RID profile, broadcast technology, authentication claim and raw message payload.

MISP Drone Object - Geolocation

Object name: `geolocation`

Link:
`https://github.com/MISP/misp-objects/blob/main/objects/geolocation/definition.json`

- Models a position with latitude, longitude and altitude in a defined spatial reference.
- Adds geographic context with address, city, region, country and ZIP code.
- Supports precision and timeline fields such as accuracy radius, EPSG, first seen and last seen.

MISP Drone Object - GPX

Object name: `gpx`

Link: `https://github.com/MISP/misp-objects/blob/main/objects/gpx/definition.json`

- Stores the GPX file itself with filename, version, creator and attachment metadata.
- Summarises temporal and spatial coverage through timestamps and WGS84 bounding coordinates.
- Quantifies the dataset with waypoint, route, track and segment counts plus file hashes.

MISP Drone Object - WiFi

Object name: `wifi-connection`

Link: `https://github.com/MISP/misp-objects/blob/main/objects/wifi-connection/definition.json`

- Identifies the wireless network with SSID and BSSID.
- Describes the radio and security setup with channel, frequency, authentication, encryption and protocol.
- Captures supporting context such as IP address, hostname, vendor, signal strength and observation times.

MISP Drone Object - Report

Object name: `report`

Link: `https://github.com/MISP/misp-objects/blob/main/objects/report/definition.json`

- Describes a report with its title, summary and external link.
- Classifies the document through report type and optional case number.
- Allows attachment of the underlying report file for sharing or archiving.

MISP Tags

Machine tags to classify drone platforms, roles and regulations

MISP Drone Tag - CE UAS Classification

Tag namespace: `ce-uas-classification`

Link: `https://github.com/MISP/misp-taxonomies/blob/main/ce-uas-classification/machinetag.json`

- Implements the EASA drone classes from C0 to C6.
- Encodes regulatory and safety properties such as weight bands, altitude limits, remote ID and geo-awareness.
- Helps align observed or seized drones with EU operational categories and compliance requirements.

MISP Drone Tag - NATO UAS Classification

Tag namespace: `nato-uas-classification`

Link: `https://github.com/MISP/misp-taxonomies/blob/main/nato-uas-classification/machinetag.json`

- Organises UAS into NATO Class I, II and III based on size and operational scale.
- Refines the classes with categories such as micro, mini, small, tactical, MALE, HALE and strike/combat.
- Supports military reporting and capability-based comparison across incidents and platforms.

MISP Drone Tag - UAS Additionnal Classification

Tag namespace: `uas-additionnal-classification`

Link: `https://github.com/MISP/misp-taxonomies/blob/main/uas-additionnal-classification/machinetag.json`

- Adds functional and technical qualifiers such as FPV, VTOL, LM, ISR, QUAD, WING, FIBER and OWA.
- Complements mass-based or regulatory taxonomies with mission profile and airframe characteristics.
- Makes it easier to tag hybrid systems and correlate shared tactics, techniques and capabilities.

MISP Galaxy

Reference knowledge base for UAV models and related metadata

MISP Galaxy - UAV Database

Name: `UAVs/UCAVs`
Link:
`https://misp-galaxy.org/uavs/`

- Official MISP Galaxy cluster for drone and UCAV knowledge, published as the `uavs` galaxy.
- The current cluster contains 739 entries with per-model descriptions, metadata and external references.

UAVs/UCAVs :: Arash-2 (Kyan 2, آرش)

Cluster ID	119466
Name	Arash-2 (Kyan 2, آرش)
Parent Galaxy	UAVs/UCAVs
Description	Arash-2 is an Iranian jet-powered long-range loitering munition designed to conduct kamikaze strikes on high-value targets, including air-defense radars and critical infrastructure. It reportedly reaches about 400 km/h, has a ~4 m wingspan, carries roughly a 30 kg warhead, and is operated by Iran for stand-off attacks.
Default	Yes
Version	25122311
UUID	5b54232d-0d7e-53fa-82eb-8b37bd74cfaa
Collection UUID	bef5c29d-b0db-4923-aa9a-809211f26d3ab
Source	OSINT
Authors	Bilal Khan, Paul Jung, Enes AYATA
Distribution	All communities
Owner Organisation	MISP
Creator Organisation	MISP
Connector tag	misp-galaxy:uavs="Arash-2 (Kyan 2, آرش)"

Tabular view | JSON view

Key ↕	Value
country	iran
operator	iran
payload_kg	30
powerplant	jet engine
refs	https://odin.tradoc.army.mil/WEG/Asset/a20576c1debc6b6cae8f8657a8b49125
refs	https://en.defence-ua.com/analysis/iranian_arash_2_kamikaze_drones_capabilities_and_specifications-4555.html
refs	https://en.wikipedia.org/wiki/Arash_(drone)
specifications	LM
top_speed_kmph	400
weapon_capability	Yes
[illegible]m	4.0

[illegible]g 11 records out of 11 total, starting on record 1, ending on 11

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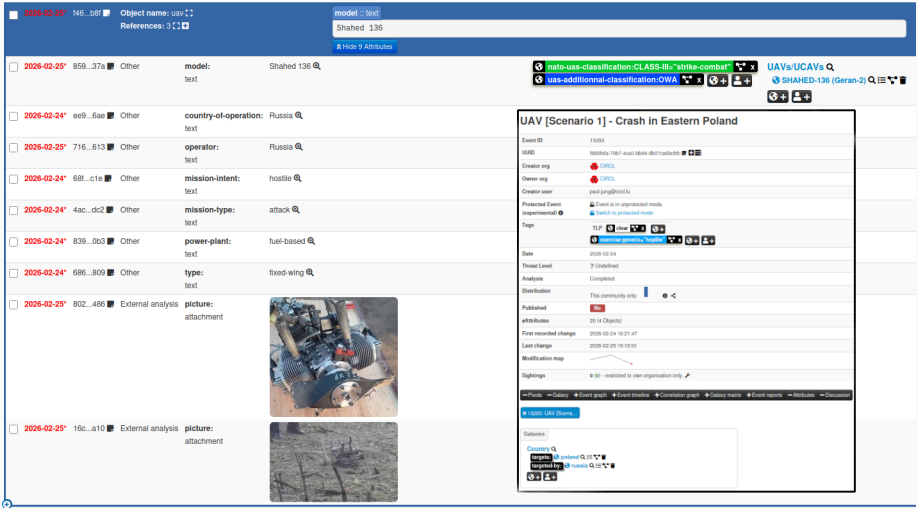
Case Study 1 – UAV Crash in Eastern Poland (Aug 2025)

- **Date:** Night of August 19–20, 2025.
- **Location:** Near the village of Osiny, Lublin region, eastern Poland (border area).
- **Platform:** Russian-made Shahed-type unmanned aerial vehicle (UAV).
- **Incident:**
  - UAV crashed and detonated.
  - Explosion caused material damage (e.g. shattered windows).
  - Debris scattered across nearby agricultural fields.
- **Preliminary assessment:**
  - Likely entered Polish airspace during ongoing regional military activity.
  - Investigation conducted by relevant authorities.
- **Event ID:** f685fefa-70b7-4ca0-bb94-db07ca6bcbfc
  - https://training5.misp-community.org/events/view/67

Case Study 1 – UAV Crash in Eastern Poland (Aug 2025)



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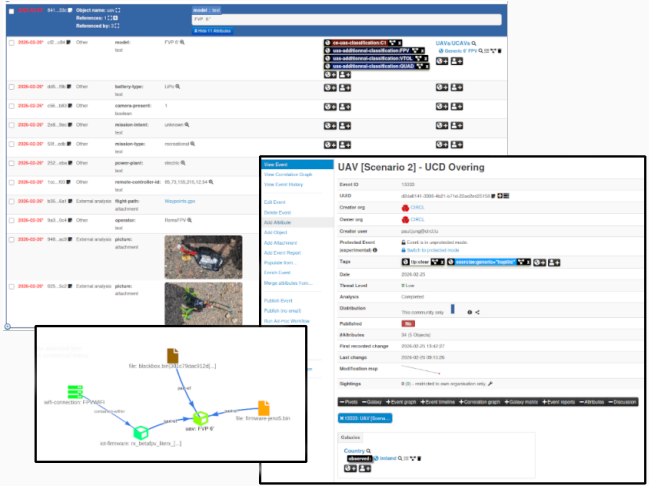
Case Study 2 – UAV over UCD (Simulation)

- **Date:** Night of December 20, 2025
- **Location:** University campus near the village
- **Platform:** Self-assembled FPV drone
- **Incident:**
  - UAV found crashed inside the premises
- **Preliminary assessment:**
  - Drone recovered; likely battery malfunction
  - Flight occurred in a restricted area
- **Event ID:** ee959409-21d1-4701-8385-fb34fedf02b9
  - https://training5.misp-community.org/events/view/69

Case Study 2 – UAV over UCD (Simulation)



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References

- https://www.misp-project.org
- https://www.misp-galaxy.org
  - https://misp-project.org/objects.html
  - https://misp-project.org/taxonomies.html
- https://github.com/misp